



Pharm*Achieve*

ASTHMA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AE Adverse effects

ASA Acetylsalicylic acid

βB Beta blocker

COX2 Cyclooxygenase-2

d Days

FEV1 Forced Expiratory Volume (in 1 second)

FVC Forced Vital Capacity

GERD Gastroesophageal Reflux Disease

HDM House Dust Mite

Hx History

ICS Inhaled Corticosteroid

ICU Intensive Care Unit

LABA Long-Acting Beta2 Agonist

LTRA Leukotriene Receptor Antagonist

MDI Metered Dose Inhaler

NSAID Non-Steroidal Anti-inflammatory Drug

OCS Oral Corticosteroid

PEF Peak Expiratory Flow

PRN As Needed

Pt Patient

SABA Short-Acting Beta2 Agonist

SAMA Short-Acting Muscarinic Agent

SLIT Sublingual Immunotherapy

Sx Symptom

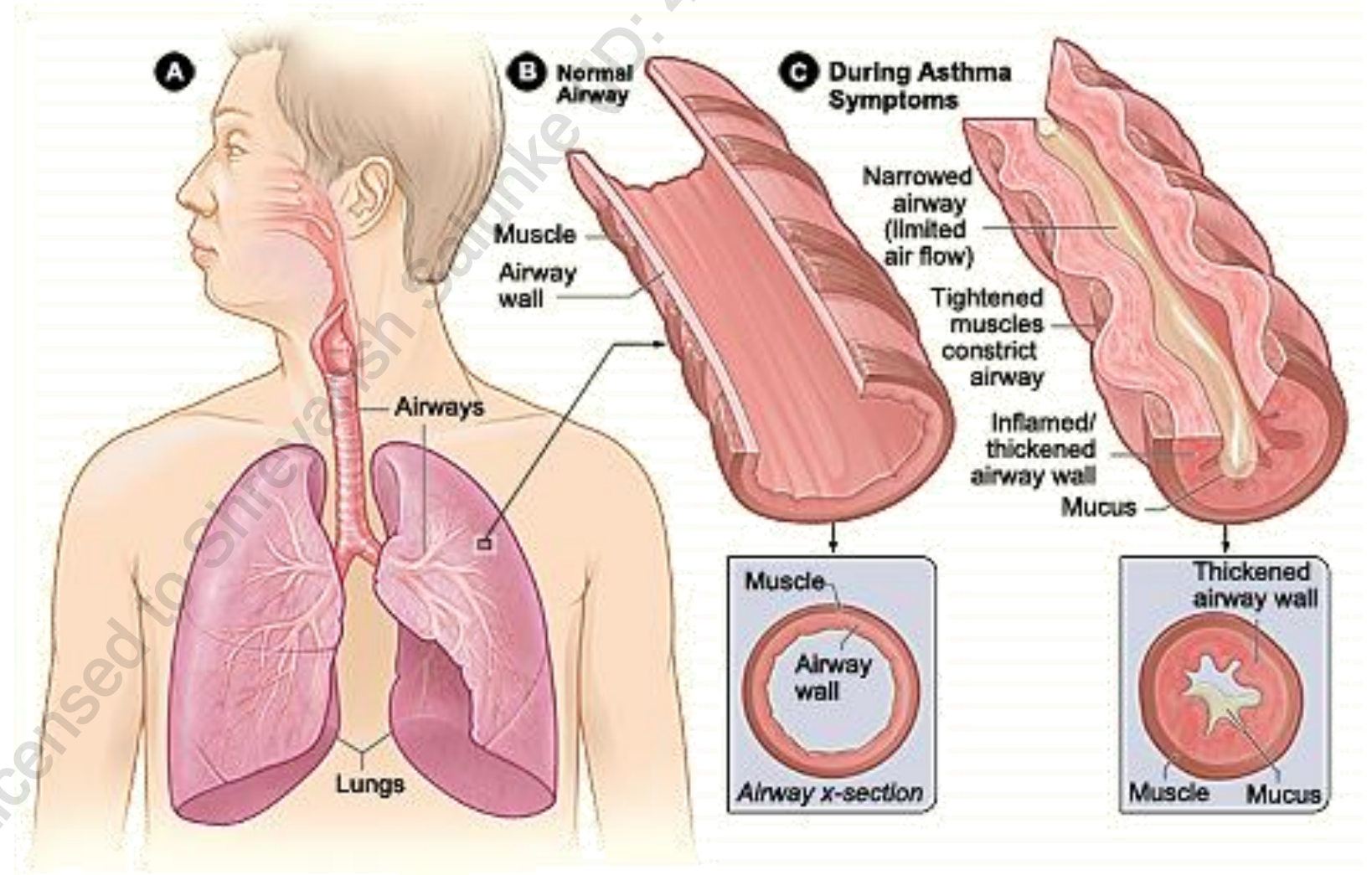
TSLP Thymic stromal lymphopoietin

Tx Treatment

Yr(s) Year(s)

PATHOPHYSIOLOGY AND CLINICAL PRESENTATION

- **Pathophysiology:** chronic inflammatory disorder of the airways resulting in airflow limitation
- **Clinical presentation:** no set pattern and variable – course tends to stabilize overtime with infrequent exacerbations
- **Symptoms:** dyspnea, chest tightening, coughing and wheezing; may be paroxysmal, intermittent or persistent (> 2 days/week)



EPIDEMIOLOGY

> 4.6 million
people in Canada have asthma

317 people
are diagnosed with asthma
every day

#1
chronic disease in childhood
(~900k patients)

40%
more prevalent in
Indigenous communities

DIAGNOSIS

- **Spirometry**: maximal inhalation followed by forceful exhalations
- **Forced Expiratory Volume (FEV₁)**: amount of air forced out in 1 second
- **Forced Vital Capacity (FVC)**: largest amount of air one can exhale after taking maximal inhalation
- **FEV₁/FVC ratio** = indicator of airway obstruction

RISK FACTORS

Asthma

- Comorbidities e.g. **obesity, atopy**
- Exposures: smoking, e-cigarettes, allergens, air pollution, occupation
- Genetics
- Impaired lung function

Potentially Poor Outcomes/Fatal Exacerbations

- ≥ 1 severe exacerbation in the previous year, hx of previous ventilation or ICU admission
- Impaired lung function
- **Excessive SABA use ($\geq 3 \times 200$ -metered dose inhalers per year)**
- Poor/declining lung function
- **ICS non-prescription/non-adherence/improper technique**
- Lower socioeconomic status
- **Medical comorbidity** (obesity, atopy, chronic rhinosinusitis, GERD, confirmed food allergy, anxiety, depression), exposure to allergens
- Pregnancy
- **Smoking and/or using e-cigarettes**

SYMPTOM/EXACERBATION PRECIPITANTS

Drug-induced

- NSAIDs, ASA
- Beta-blockers
- Sulfites
- Benzodiazepines
- Opioids
- Tartrazine

Psychosocial/behavioural

- Non-adherence
- Smoking
- Intense emotion
- Anxiety
- Laughter

Other

- GERD
- Allergen exposure
- Cold exposure

GOALS OF THERAPY

- ✓ Achieve and maintain long-term control of disease:
 - ✓ Prevent daytime and nocturnal symptoms
 - ✓ Maintain regular activities
 - ✓ Prevent need for reliever therapy
- ✓ Prevent exacerbations and deaths related to asthma
- ✓ Administer effective pharmacotherapy while minimizing adverse effects

DEFINITION OF GOOD ASTHMA CONTROL

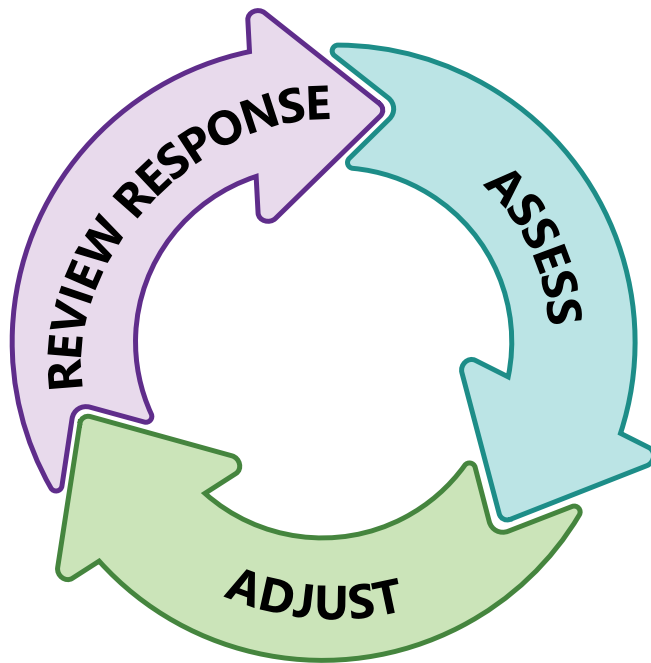
Parameter	Frequency/Severity (assessed over the past 4 weeks)
Daytime symptoms	≤2 days/week
Nighttime symptoms (e.g. night waking)	None
Physical Activity	Normal (no limitation)
Exacerbations	Mild and infrequent
Absence from school/work	None
Reliever Use (SABA or bud/form*)	≤2 doses/week



*No established criteria for control using budesonide/formoterol as reliever therapy, but frequency of use should still be assessed

PERSONALIZED ASTHMA MANAGEMENT

ASSESS, ADJUST, AND REVIEW RESPONSE



Assess	Review diagnosis, symptom control , modifiable risk factors (including lung function), comorbidities, inhaler technique and adherence , patient preferences and goals
Adjust	Treat modifiable risk factors and comorbidities, incorporate non-pharmacological measures and drug measures, educate patients
Review Response	Review symptoms, exacerbations, side effects, lung function, comorbidities, consider biomarkers, patient satisfaction

NON-PHARMACOLOGICAL MEASURES

- Avoid precipitating factors (e.g. occupational irritants, indoor and outdoor environmental allergens/pollutants if possible)
- Smoking cessation
- Encourage physical activity
- Weight loss if patient is obese
- Stress management techniques
- Treat modifiable co-morbidities (e.g. obesity, anxiety, depression, GERD)
- Hyposensitization therapy to allergens
 - Note: inadequate evidence currently available to support using home air cleaners/purifiers to reduce allergens
- Provide asthma education and develop a written action plan for all patients

ASTHMA ACTION PLAN

Education Plan	Written Action Plan
Asthma Pathophysiology	How/When to assess control
Triggers	Controller adherence
"Good Control" parameters	Environmental changes
Exacerbation risk factors	Management of worsening symptoms
Reliever vs. control	When to seek help
Medication Safety/ADR	
Inhaler Technique	

PHARMACOLOGICAL MEASURES - OVERVIEW OF INHALERS

Medication Class	Beta-2 Agonists	Muscarinic Antagonists	Inhaled Corticosteroids
Medications in Class	Salbutamol, terbutaline, formoterol, indacaterol, vilanterol, and salmeterol	Tiotropium, ipratropium, aclidinium, glycopyrronium, umeclidinium	Beclomethasone, budesonide, ciclesonide, fluticasone propionate and furoate, and mometasone
Mechanisms of Action	- Beta-2 receptors - Found in many parts of the body including the heart - Primarily present in the smooth muscles of the airways. - Agonism or binding and activation of the beta-2 receptors mimicking the actions of epinephrine or adrenaline - Epinephrine or adrenaline triggers a signalling cascade = smooth muscle relaxation and bronchodilation	- Muscarinic receptors respond to acetylcholine - Acetylcholine binding increases levels of bronchial mucus + bronchoconstriction - Muscarinic antagonists = prevent constriction and reduce mucus production	- Inflammation in the lungs limits airflow - Corticosteroids = reduce the number of inflammatory cells - Additionally act on: - Structural cells = reduce cell leakiness, increase the concentration of beta-2 receptors to increase SABA and LABA effectiveness - Mucus glands = reduce mucus secretion
Side Effects	Tachycardia, palpitations, restlessness, tremors, and insomnia (more so if the medication is high dosed and swallowed) Indacaterol is associated with brief cough and arthralgias.	Dry mouth, urinary retention and metallic taste are common Aclidinium has the lowest incidence compared to tiotropium and glycopyrronium	Oral thrush (advise patients to rinse mouth after use to reduce risk) Dysphonia or hoarse throat Higher doses could worsen intra-ocular pressure and increase pneumonia risks

PHARMACOLOGICAL MEASURES

Reliever Medications

- Low dose ICS-formoterol: budesonide/formoterol
- Short Acting Beta Agonists (SABA): salbutamol, terbutaline
- Short Acting Muscarinic Antagonists (SAMA): ipratropium

Controller Medications

- Inhaled Corticosteroids (ICS): e.g., fluticasone
- Long-acting Beta Agonists (LABA): e.g. salmeterol
- Leukotriene Receptor Antagonists (LTRA): e.g. montelukast
- Long-acting Muscarinic Antagonists (LAMA): e.g. tiotropium

Other Medications

- Systemic corticosteroids: e.g. prednisone
- Methylxanthines: e.g. theophylline
- Biologics: e.g. dupilumab, omalizumab

PHARMACOLOGICAL MEASURES

Anti-Inflammatory Reliever (AIR)

- ICS-formoterol
 - Budesonide/Formoterol PRN
 - **Note:** mometasone/formoterol is licensed in Canada for maintenance of asthma (ICS/LABA combination), but is **NOT** currently approved for AIR

Maintenance And Reliever Therapy (MART)

- Routine daily use of budesonide/formoterol + PRN reliever use of budesonide/formoterol

Budesonide/formoterol is only Health Canada-approved for children and adults 12 years and older!

WHY IS SABA MONOTHERAPY NO LONGER RECOMMENDED?

Asthma is not only a disease of bronchoconstriction

Regular use increases allergic responses and airway inflammation

SABA monotherapy is associated with ↑ risk of exacerbations, ↓ lung function and asthma-related deaths

PRN budesonide/formoterol combination was superior to SABA monotherapy PRN with regular budesonide monotherapy in reducing severe exacerbations along with overall steroid exposure in mild asthma

Initiating treatment with SABA monotherapy reinforces the idea that it is the primary therapy, which increases risk of future non-compliance to prescribed ICS-containing regimens

STEP-UP AND STEP-DOWN THERAPY

Step Up Therapy

Consider stepping up therapy when current therapy is **not providing sufficient asthma control** after 2-3 months

- First ensure:
 - Adherence is optimized
 - Triggers are removed
 - Inhaler technique optimized

Step Down Therapy

Consider stepping down therapy **once asthma is under total control** and lung function is stable (no exacerbations, no night awakenings, daytime symptoms twice or less per week) **in 3-month intervals** (every 3-6 months)

- **Do not consider step-down if** the patient is pregnant, traveling to a different climate, or experiencing a respiratory infection
- Usually, **reduce ICS dose by 25-50%**. Stopping LABA may lead to deterioration
- May need to add a LABA to reduce the dose of ICS
- **Do not stop ICS-containing treatment**

LONG-ACTING INHALED BETA₂ AGONISTS (LABA)

- Examples: salmeterol, formoterol, indacaterol, vilanterol
- Only use in patients already taking an ICS
 - May be ICS-sparing (consider reducing the dose of ICS)
 - Regularly using LABA monotherapy is associated with an increased risk of death in patients with asthma
- In individuals using budesonide/formoterol as the ICS/LABA combination therapy, it may be used both as reliever and controller therapy due to formoterol's rapid onset of action
- Salmeterol, vilanterol, indacaterol have a slow onset of action: DO NOT USE AS RELIEVER

TREATMENT ALGORITHM

ADULTS AND ADOLESCENTS (12 YEARS AND OLDER)

Step 1 & 2	Symptoms less than 3 to 5 days/week AND normal (or mildly reduced) lung function Low-dose ICS-formoterol prn (preferred)	Step 1	Symptoms for 1-2 days/week or less: Alternative: Low-dose ICS taken whenever SABA prn is taken
		Step 2	Alternative: Low-dose maintenance ICS + ICS-SABA prn or SABA prn
Step 3	Symptoms most days of the week OR nocturnal awakenings ≥1 time/week OR low lung function - Low-dose MART with ICS-formoterol - Alternatives: Low-dose ICS-LABA maintenance + ICS-SABA PRN or SABA PRN		
Step 4	Daily symptoms, nocturnal awakenings ≥1 time/week AND low lung function OR recent exacerbation - Medium-dose MART with ICS-formoterol - Alternatives: Medium dose ICS-LABA maintenance + ICS-SABA PRN or SABA PRN		
Step 5	Uncontrolled symptoms despite Step 4 management - Refer To Specialist - Add-on LAMA with low dose ICS-formoterol PRN (reliever) - Consider: high dose ICS-formoterol maintenance +/- anti-IgE, anti-IL5/5R, anti-IL4Rα, anti-TSLP - Alternative reliever: ICS-SABA PRN or SABA PRN		

TREATMENT ALGORITHM

CHILDREN (AGED 6-11 YEARS OLD, INCLUSIVE)

Step 1

Symptoms occur **less than 1 to 2 days per week**

- Low-dose ICS taken whenever SABA PRN is taken

Step 2

Symptoms **2 to 5 days per week**

- Daily low dose ICS with SABA PRN
- **Alternative:** LTRA OR Low-dose ICS taken whenever SABA PRN is taken

Step 3

Symptoms most days of the week (4-5 days/week) **OR** nocturnal awakenings **≥1 time/week**

- Low dose ICS-LABA (+SABA PRN) or medium-dose ICS (+SABA PRN) or very low-dose ICS-formoterol MART
- **Alternative:** Low-dose ICS and LTRA + SABA PRN

Step 4 *

Daily asthma symptoms **AND** nocturnal awakenings **≥1 time/week AND** low lung function

- Medium-dose ICS-LABA daily (+SABA PRN) or low-dose ICS-formoterol MART
- Consider referring for expert advice
- **Alternative:** add LTRA or tiotropium

* If symptoms persist despite step 4 treatment, patient may require referral for phenotypic assessment

ALTERNATIVE THERAPIES

FYI

Short-acting muscarinic antagonists (or anticholinergics)
e.g. ipratropium

- Can be considered as add-on to β_2 -agonists for severe acute exacerbations
- May also be an alternative for those susceptible to tremor or tachycardia from β_2 -agonists
- The onset of ipratropium is delayed compared with β_2 -agonists however the bronchodilator effects last longer
- May also consider in beta-blocker-induced bronchospasm

Long-acting muscarinic antagonists (or anticholinergics)
e.g. tiotropium

May be used as add-on therapy in patients symptomatic despite using maintenance treatment e.g. high-dose ICS/LABA, medium-dose ICS, low-dose ICS, and medium-dose ICS+/-LABA

ALTERNATIVE THERAPIES

Leukotriene receptor antagonists (LTRA) e.g. montelukast PO

- Have anti-inflammatory properties however they are less effective than low-dose ICS for improving symptoms or preventing exacerbations
- As a result, they are considered 2nd line monotherapy
- May be consider as add-on to low-dose ICS in patients with concomitant rhinitis even though ICS/LABA is more effective than ICS+LTRA in preventing exacerbations
- Side effects: headache, flu-like symptoms, abdominal pain, nightmares, and **neuropsychiatric effects** (e.g. depression)

Methylxanthines e.g. theophylline PO, aminophylline IV – **FYI**

Not recommended due to its side effect profile and lower efficacy, **narrow therapeutic index**/toxicity, drug interactions, and due to the requirement of monitoring blood levels

BIOLOGICS

FYI

Medication Class	Drug name	Indication	Adverse effects
IgE-neutralizing antibody	Xolair® (omalizumab) → reference biologic Omlyclo® (omalizumab) → biosimilar	May be considered as add-on for patients with severe allergic asthma not controlled with high dose ICS-LABA	↑ risk of infection, anaphylaxis (first three doses given in medical setting), headache, fatigue, injection site reactions
Interleukin-5 inhibitors	Nucala® (mepolizumab) Fasenra® (benralizumab) Cinqair® (reslizumab)	Add-on therapy for adults with severe eosinophilic asthma who are poorly controlled on high dose ICS-LABA	↑ risk of infection, fatigue, headache injection site reactions
Interleukin-4 & 13 inhibitors	Dupixent® (dupilumab)	Approved as adjunctive treatment of moderate-severe eosinophilic asthma Improves lung function, reduces annual incidence of severe exacerbations	↑ risk of infection, fatigue, conjunctivitis, injection site reactions
Anti-thymic stromal lymphopoietin (anti-TSLP)	Tezspire® (tezepelumab)	Add-on maintenance for severe asthma	↑ risk of infection, fatigue, sore throat, injection site reactions

ASTHMA EXACERBATIONS

Asthma exacerbation: an acute flare-up of asthma that may be triggered by an external irritant or allergen, the environment (e.g. cold air), or exercise

- Depending on the severity of the exacerbation, patients may require hospitalization

Asthma action plans help patients to manage potential exacerbations

Green Zone

Asthma is under control

Yellow Zone

Asthma is getting worse

- Difficulty exercising, difficulty maintaining sleep, increased frequency of reliever inhaler, absences from work/school
- Should book an appointment with family physician to have medications reviewed

Red Zone

Asthma emergency

- Chest tightness, use of accessory muscles to support breathing, sweating (premonitory sign of severe exacerbation), excessive coughing or wheezing, gasping voice, difficulty talking, cyanosis, reliever medication is not providing relief
- Need to seek immediate medical attention

EXACERBATION MANAGEMENT

YELLOW ZONE MANAGEMENT

Age 6-11

- SABA PRN
- +/- systemic prednisone/prednisolone: continue for 3-5 days total
- Maintain $O_2 \geq 94\%$
- Arrange follow-up visit within 1-3 days
- Increase maintenance dose of ICS or ICS-LABA for next 2-4 weeks
- Antibiotics are NOT recommended (unless fever, purulent sputum or radiographic evidence of pneumonia)



Note: SABA onset is 5-15 minutes, and duration of action is 3-6 hours

EXACERBATION MANAGEMENT

YELLOW ZONE MANAGEMENT

Age $\geq$ 12

- SABA PRN
- +/- systemic prednisone/prednisolone
- Maintain O₂ at 93-95%
- Arrange follow-up visit within 2-7 days
- Increase maintenance dose of ICS or ICS-LABA for next 2-4 weeks
- Antibiotics are NOT recommended (unless fever, purulent sputum or radiographic evidence of pneumonia)

EXACERBATION MANAGEMENT

RED ZONE MANAGEMENT

All ages

- Patients must be referred to the emergency department
- SABA +/- SAMA standing + prn (high dose)
- Oxygen
- Systemic corticosteroids
- MgSO_4 or heliox for salvage therapy
- Respiratory support



REASSESS asthma therapy, adherence and precipitants/risk factors!!!

SEVERITY ASSESSMENT

Patients should be on therapy for at least 2-3 months before assigning a severity

Severity	Definition
Mild	Well-controlled using low intensity regimens: PRN low-dose ICS- formoterol OR Low-dose ICS plus SABA PRN
Moderate	Well-controlled using step 3 or 4 regimens: Low/medium-dose ICS-formoterol MART OR Low/medium-dose ICS-LABA + SABA PRN
Severe	Requires high-dose ICS-LABA to remain controlled OR uncontrolled despite high-dose ICS-LABA

SEVERITY ASSESSMENT

FYI

Severity	GINA 2025 Guidelines	CTS 2021 Guidelines
	Controlled On:	Controlled On:
Mild	Well-controlled using low intensity regimens: PRN low-dose ICS- formoterol OR Low-dose ICS plus SABA PRN	Low-dose ICS (or LTRA) plus SABA PRN OR PRN budesonide-formoterol
Moderate	Well-controlled using step 3 or 4 regimens: Low/medium-dose ICS-formoterol MART OR Low/medium-dose ICS-LABA	Low-dose ICS + second controller plus SABA PRN OR Moderate dose ICS +/- second controller plus SABA PRN OR Low/moderate-dose budesonide-formoterol plus budesonide-formoterol PRN
Severe	Requires high-dose ICS-LABA to remain controlled OR uncontrolled despite high-dose ICS-LABA	High dose ICS + second controller in past year OR Systemic corticosteroids for 50% of the past year OR Remains uncontrolled with therapy

SPECIAL POPULATIONS - PREGNANCY AND BREASTFEEDING

- Inadequate asthma control during pregnancy is associated with worse outcomes (preterm birth, low birth weight, congenital anomalies, pre-eclampsia, placenta previa)
- **DO NOT STEP-DOWN THERAPY DURING PREGNANCY!**
- Follow-up should be done **every 4-6 weeks** in pregnancy to review asthma control

Medication	Considered Safe in Pregnancy	Considered Safe in Breastfeeding
SABA	Yes	Yes
LABA	Some data suggests so	Yes
ICS	Yes (especially budesonide)	Yes
LAMA/SAMA	Yes, if needed	Yes
Theophylline	Yes	Yes
Montelukast	Yes	Transfers into breastmilk
Biologics	Not enough data, avoid	Not enough data, avoid

SPECIAL POPULATIONS

FYI

- Patients ≤ 5 years old
 - 2025 GINA update clarified that asthma CAN be diagnosed in this age group
 - Diagnosis based on the presence of ALL 3 criteria (otherwise 'suspected asthma'):
 - ≥ 2 acute wheezing episodes in last 12 months OR one wheezing episode with asthma-like symptoms (e.g. dry cough, symptoms worse after exertion)
 - No other diagnosis explains respiratory symptoms
 - Fast clinical response to SABA at home or medical facility OR improvement of asthma symptoms and/or wheezing episodes with daily ICS trialed for 2-3 months
- Patients with very severe asthma unresponsive to the maximum stepwise approach
 - Assessment of asthma phenotype and consideration of biologic therapy
- Patients with exercise-induced asthma
 - Pre-exercise treatment with AIR effective (Alternative ICS with SABA prn OR as detailed in stepwise pharmacotherapy approach)

VACCINATIONS

- Patients with asthma are at an increased risk of complications from many respiratory viruses and should be counseled on receiving appropriate immunizations
 - Influenza
 - COVID-19
 - RSV
 - Invasive Pneumococcal Disease (IPD)
 - Especially if required hospitalization in the previous 12 months
- Always review vaccination status during the start of respiratory virus season

APPENDIX: ICS DOSING (as per CTS)

FYI

Drug	CHILDREN			ADULTS		
	Low dose	Medium dose	High dose	Low dose	Medium dose	High dose
Beclomethasone	$\leq 200\mu\text{g/d}$	200-400 $\mu\text{g/d}$	$> 400\mu\text{g/d}$	$< 200\mu\text{g/d}$	200-500 $\mu\text{g/d}$	$> 500\mu\text{g/d}$
Fluticasone propionate	$\leq 200\mu\text{g/d}$	$> 200\text{--}400\mu\text{g/d}$	$> 400\mu\text{g/d}$	$< 250\mu\text{g/d}$	250-500 $\mu\text{g/d}$	$> 500\mu\text{g/d}$
Ciclesonide	$\leq 200\mu\text{g/d}$	200-400 $\mu\text{g/d}$	$> 400\mu\text{g/d}$	$< 200\mu\text{g/d}$	200-400 $\mu\text{g/d}$	$> 400\mu\text{g/d}$
Budesonide	$\leq 400\mu\text{g/d}$	400-800 $\mu\text{g/d}$	$> 800\mu\text{g/d}$	$< 400\mu\text{g/d}$	400-800 $\mu\text{g/d}$	$> 800\mu\text{g/d}$
Mometasone	$< 100\mu\text{g/d}$	200-400 $\mu\text{g/d}$	$> 400\mu\text{g/d}$	100-200 $\mu\text{g/d}$	200-400 $\mu\text{g/d}$	$> 400\mu\text{g/d}$



ICS may reduce airway remodeling

APPENDIX: RECOMMENDED INHALER DEVICES BY AGE

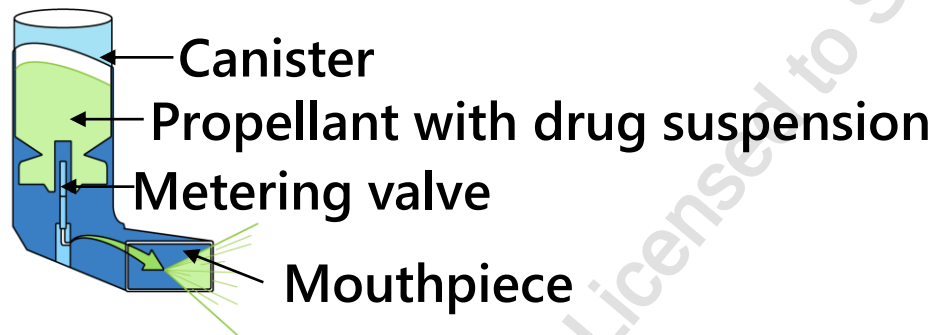
FYI

Age (years)	Device Type
0-3	MDI + masked valved holding chamber
4-6	MDI + mouthpiece valved holding chamber
> 6-11	DPI or MDI + mouthpiece valved holding chamber
≥ 12	DPI or MDI + mouthpiece valved holding chamber

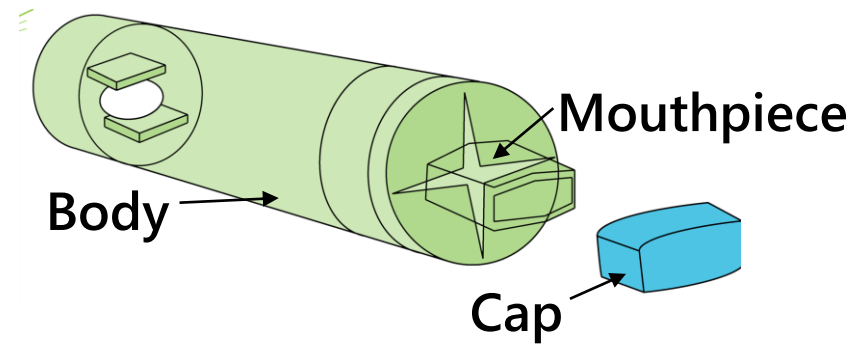
APPENDIX: MDI INHALER ADMINISTRATION

1. Shake the inhaler
2. Remove the cap
3. If using for first time or first time in a while, prime the inhaler
4. Breathe out, away from the inhaler (or spacer if used)
5. Place the inhaler (or spacer if used) in your mouth
6. Breathe in slowly and at the same time, press the top of the inhaler
7. Remove the inhaler from your mouth and try to hold breath for about 10 seconds
8. Breathe out

Inhaler



Aerochamber (spacer)



APPENDIX: DPI ADMINISTRATION (GENERAL)

1. Remove or unscrew the mouthpiece cover (if applicable)
2. Load a dose
3. Do not invert, tilt or shake the inhaler
4. Breathe out, away from the inhaler
5. Place the inhaler in your mouth
6. Breathe in deeply and strongly to deposit drug into the lung
7. Remove the inhaler from your mouth and try to hold breath for about 10 seconds
8. Breathe out slowly
9. Wipe mouthpiece with a dry cloth or tissue, replace cap and store
(DO NOT USE WATER)
10. If steroid containing inhaler used, rinse mouth

MONITORING PARAMETERS: EFFICACY ENDPOINTS

Parameter	Degree of Change	Timeframe
Daytime respiratory symptoms (wheezing, dyspnea, chest tightness, cough)	Reduce to ≤ 2 times per week	Patient/parent: daily Physician: Baseline, then 1-3 months after initiating therapy, then every 3-12 months (frequency depends on several factors e.g. symptom control, risk factors, response to treatment)
Nighttime respiratory symptoms (awakening due to wheezing, dyspnea, chest tightness, cough)	Eradicate	Patient/parent: nightly Physician: see above
Spirometry (FEV1) or peak expiratory flow	Improved or stable personal best	Physician: see above
Frequency of SABA use	Reduce to ≤ 2 doses per week and < 3 canisters/year	Physician: see above Pharmacist: at every refill
Activity limitation due to asthma symptoms	Eradicate	Physician: see above
Inhaler technique	Optimal	Physician: see above Pharmacist: at every refill
Exacerbation history	None	After an exacerbation, review within 1 week

MONITORING PARAMETERS: SAFETY ENDPOINTS

Parameter	Degree of Change	Timeframe
Height (for children)	Expected growth	Annually
Inhaler technique	Optimal If using ICS: ensure mouth is rinsed after use	Physician: annually Pharmacist: at every refill
Tremor and/or tachycardia	None or tolerable	After 1-2 weeks of use (tolerance)
Neuropsychiatric effects (mood or behavioural changes, suicidal ideation)	None	Patient/parent: daily Physician: at every follow-up
Oropharyngeal candidiasis	Minimize	Pharmacist: at every refill

TAKEAWAY

- Asthma is a common chronic inflammatory disorder characterized by airflow limitation
- Common symptoms include dyspnea, chest tightness, and wheezing – these symptoms often worsen after exercise or other physical exertion
- Non-pharmacologic measures should be utilized to reduce exposure to known triggers and to educate patients on the disease state
- An asthma action plan (detailing the steps to follow in the event of an exacerbation) should be made
- Asthma treatment follows a stepwise approach based on asthma symptom severity and frequency, including frequency of symptoms during the day, any nighttime awakenings, use of SABA/reliever therapy, and missed days of school/work
- Patients may be stepped up or down based on severity; pregnant patients should NEVER be stepped down

CASE 1

JM, an 8-year-old boy, presents to your family health team practice accompanied by his mother. He is experiencing a cough and wheezing which woke him up last night and which continued into today. His mother states that he has had similar episodes in the past, the most recent being 3 weeks ago. He has stable vital signs but is experiencing symptoms of bronchospasm and is presumptively diagnosed with asthma pending further evaluation. JM has seasonal allergic rhinitis for which he takes loratadine 5 mg PO daily. He is also allergic to pet dander. He lives with his parents; his father is a smoker.

1. In which severity criteria of asthma does JM currently fit?
 - a) Very mild
 - b) Mild
 - c) Moderate
 - d) Unable to assess



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2. Of the following options, which is the MOST appropriate for JM at this time?

- a) Budesonide/formoterol 100 mcg/6 mcg 1 inhalation BID PRN
- b) Fluticasone 50mcg 1 puff inhaled BID + salbutamol 100 mcg/actuation 2 inhalations Q4H PRN
- c) Montelukast 5mg PO QHS
- d) Fluticasone 100 mcg/salmeterol 25 mcg 1 puff inhaled BID + salbutamol 100mcg/puff, 2 puffs Q4H PRN with spacer device



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3. Which of the following would indicate good asthma control for JM?

- a) No nocturnal awakenings due to asthma symptoms
- b) Salbutamol use once every other day
- c) Less than two hospitalizations for asthma per year
- d) Less than two days of school missed per month due to asthma symptoms



CASE 1

JM, an 8-year-old boy, presents to your family health team practice accompanied by his mother. He is experiencing a cough and wheezing which woke him up last night and which continued into today. His mother states that he has had similar episodes in the past, the most recent being 3 weeks ago. He has stable vital signs but is experiencing symptoms of bronchospasm and is presumptively diagnosed with asthma pending further evaluation. JM has seasonal allergic rhinitis for which he takes loratadine 5 mg PO daily. He is also allergic to pet dander. He lives with his parents; his father is a smoker.

4. Which of the following would be the most appropriate inhaler device for JM?

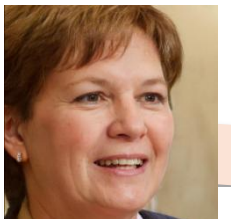
- a) Pressurized metered dose inhaler with face mask aerochamber
- b) Pressurized metered dose inhaler with mouthpiece aerochamber
- c) Dry powder inhaler
- d) Nebulizer



CASE 2

WW is a 55-year-old female with very mild asthma. WW's medical history is significant for asthma and atopic dermatitis, for which she takes salbutamol 100 mcg/actuation inhale 1-2 puffs Q4-6H PRN and betamethasone 0.1% applied to affected areas PRN. WW typically uses her salbutamol inhaler 4-5 times per month, more frequently during the summer. She drinks 1 glass of wine occasionally with dinner and smokes 5 cigarettes/day.

1. WW presents to your community pharmacy seeking a refill for her salbutamol inhaler. Today is March 1st 2026, she last filled her inhaler in April of 2025. Which of the following would be most appropriate?
 - a) Advise WW that based on her refill history, you are concerned that she is using her inhaler too frequently. Overuse of SABA inhalers is a sign of poorly controlled asthma. Refuse to fill her prescription until she visits her family physician for reassessment
 - b) Discuss the benefits of low-dose ICS/formoterol PRN. Ask her permission to fax her doctor on her behalf to request they switch her to low-dose ICS/formoterol PRN
 - c) Discuss the benefits of low-dose MART with ICS/formoterol. Ask her permission to fax her doctor on her behalf to request they switch her to low-dose MART with ICS/formoterol
 - d) Refill WW's salbutamol prescription as requested



CASE 2

WW is a 55-year-old female with very mild asthma. WW's medical history is significant for asthma and atopic dermatitis, for which she takes salbutamol 100 mcg/actuation inhale 1-2 puffs Q4-6H PRN and betamethasone 0.1% applied to affected areas PRN. WW typically uses her salbutamol inhaler 4-5 times per month, more frequently during the summer. She drinks 1 glass of wine occasionally with dinner and smokes 5 cigarettes/day. Her BMI is 27 kg/m².

2. Which of the following recommendations would NOT be appropriate for WW?
- a) Smoking cessation
 - b) Avoidance of indoor and outdoor allergens/pollutants
 - c) Avoidance of physical exertion that may trigger asthma symptoms
 - d) Weight loss



CASE 2

WW is a 55-year-old female with very mild asthma. WW's medical history is significant for asthma and atopic dermatitis, for which she takes salbutamol 100 mcg/actuation inhale 1-2 puffs Q4-6H PRN and betamethasone 0.1% applied to affected areas PRN. WW typically uses her salbutamol inhaler 4-5 times per month, more frequently during the summer. She drinks 1 glass of wine occasionally with dinner and smokes 5 cigarettes/day.

3. Now that WW's drug therapy includes an ICS, should she stop using topical betamethasone?

- a) Yes, as she is at high risk of corticosteroid toxicity and associated side effects
- b) Yes, she should be switched to a lower potency topical corticosteroid
- c) No, as she is at low risk of corticosteroid toxicity and associated side effects
- d) Yes, if she requires daily ICS treatment



CASE 2

WW returns to your pharmacy 3 years later. She was stable on low-dose ICS/formoterol used when needed for the first 2 years, then her symptoms progressed and her family physician stepped-up her therapy to low-dose MART with ICS/formoterol last year. Over the past few weeks, WW has been experiencing increased dyspnea and a cough that is sometimes productive. She finds that her symptoms worsen at night and with physical exertion. WW has attempted to quit smoking multiple times over the past few years, but unfortunately, was never successful.

4. Which of the following recommendations would be most appropriate?

- a) Refer WW for a lung function test to assess for comorbid COPD
- b) Step-up her therapy to high-dose MART with ICS/formoterol
- c) Switch her to medium-dose maintenance therapy with ICS/LABA + SABA PRN
- d) Refer WW for a chest x-ray to assess for pneumonia



REFERENCES

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CHANGE LOG

June 2025

- Slide 29: Updated asthma severity to match GINA guidelines
- Slide 30: New slide comparing GINA vs CTS for asthma severity
- Slide 42: Changed answer to case question to D based on slide 29

July 2025

- Slide 11: Added 'consider biomarkers' based on 2025 GINA update
- Slide 21: Changed medium/high dose ICS to medium dose ICS in step 4 alternative treatment as per 2025 GINA update
- Slide 29: Severity assessment updated based on GINA
- Slide 30: New
- Slide 32: Special populations (children ≤ 5 yrs of age) updated based on GINA 2025

October 2025

- Slide 7: Defined excessive SABA use
- Slide 10: Changed FEV value to $\geq 80\%$ predicted per GINA guidelines and added side note
- Slide 31: Consolidated information into table
- Slide 34: Updated takeaway

March 2026

- Slide 5: Updated statistics
- Slide 7: Updated risk factors
- Slide 14: Added metallic taste to muscarinic antagonists, clarified that acetylcholine *increases* mucus production
- Slide 12: Added physical activity, weight loss and stress management per GINA guideline
- Slide 17: Updated slide on SABA monotherapy
- Slide 18: Changed timeline to step-up to 2-3 months, added "reduce ICS dose by 25-50%", added "stopping LABA may lead to deterioration"

- Slide 21: changed step 4 to "daily" asthma symptoms per GINA
- Slide 23: specified that montelukast is less effective than low-dose ICS
- Slide 24: Added slide on biologics
- Slides 26,27,28: Updated to closely match GINA 2025
- Slide 29: Updated definition of severe asthma
- Slide 33: Added RSV vaccine
- Slide 35: Switched from 1-3 to 0-3 for face mask recommendation
- Slide 38-39: Added monitoring parameters
- Slide 40: Made more concise
- Slide 45-48: Edited case 2 and questions

March 2026

- Slide 10: Updated wording of nighttime symptoms to match GINA guidelines